FOLD
Fourier Overlap Linguistic Distance

Thom Lazor

Code

- https://github.com/tlazor/fold
- GitHub - facebookresearch/XNLI: Evaluating Cross-lingual Sentence Representations
    - 15 languages, sentences translated side by side
- multilingual.md - google-research/bert · GitHub
    - Multilingual bert (~102 languages)

Baselines

- Full article: Mutual intelligibility between closely related languages in Europe
  - Germanic, Slavic and Romance families analyzed for mutual intelligence
  - Asymmetry observed (everyone was better at english than english speakers were for other Germanic languages)
  - Results are also compiled for listeners with minimal exposure to the speaker’s language
- Foreign Language Training - United States Department of State
  - Data on how long US diplomats are estimated to take learning a number of languages

Related Workshop

The Twelfth Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial) 2025 | ACL Member Portal

Potential Dataset: Shared Tasks - VarDial 2023

- Corpus of English (American and British), Portuguese (Brazilian and European), Spanish (Argentinian and Peninsular)

Related Work

- Using relative entropy for detection and analysis of periods of diachronic linguistic change
  - Detection of linguistic change in certain periods of scientific english (1700-1840) using KL divergence
  - Lexical - unigrams; Grammatical - PoS trigram
  - Sliding window comparing periods before and after given year to calculate entropy (large change indicates linguistic change)
  - Interesting graphs showing most significant unigram/trigram differences between pre and post periods
  - code: https://stefaniadegaetano.com/code/
  - Similar paper using KL divergence and trigrams for distance analysis: Information Theory and Linguistic Variation: A Study of Brazilian and European Portuguese

Related Work

- https://aclanthology.org/W16-1208.pdf
  - Jensen-Shannon divergence between cosine similarity distributions of paired word vectors
  - Used bible to compare 50 natural languages
- From Language Identification to Language Distance
  - Perplexity of language A n-gram model on test set of language B (note the asymmetry)
  - Compares 44 languages
  - Similar work for historical varieties: Measuring language distance among historical varieties using perplexity. Application to European Portuguese.

Ideas

- Mean Square Average of cosine distance (embeddings) or L1 distance (likelihood)
  - Imagine the tokens as tracing a path in n-dim space to the final meaning, which should be similar regardless of language (maybe)
  - Large differences in path could mean large linguistic differences
  - symmetric
- Overlap of Fourier Power Spectra (likelihood/embeddings)
  - High overlap means similar variance at given frequencies
  - Symmetric, so may not capture asymmetric distance differences that can be significant
- KL Divergence of Power Spectra (likelihood/embeddings)
  - Asymmetric